

EXPERIMENT NO. 6 — HIERARCHICAL CLUSTERING ALGORITHM

Learning Objective: Student should be able to implement a Hierarchical Clustering algorithm.

Tools: Java / C / C++ / Python / Data Mining Tools

Theory: Implementation of Clustering Algorithm (Hierarchical).

1. Cluster Analysis

Cluster analysis, or clustering, is an **unsupervised learning** technique used to divide a collection of observations into groups called clusters. The objective is to place observations that are similar to one another in the same cluster while keeping dissimilar observations in different clusters.

Unlike classification, clustering does not require predefined class labels. The algorithm discovers the grouping structure from the input data itself. Clustering is widely used in data mining, machine learning, pattern recognition, customer segmentation, image analysis, bioinformatics and exploratory data analysis.

2. Hierarchical Clustering

Hierarchical clustering creates a hierarchy of clusters rather than directly producing only one fixed partition. The hierarchy is commonly represented by a **dendrogram**, a tree-like diagram that shows the order and distance at which clusters are merged or divided.

An important advantage is that the user does not necessarily need to specify the number of clusters before the hierarchy is constructed. A desired number of clusters can later be obtained by cutting the dendrogram at an appropriate level.

3. Types of Hierarchical Clustering

Type	Description
Agglomerative (Bottom-Up)	Each observation starts as its own cluster. The closest clusters are repeatedly merged until the desired hierarchy is obtained.
Divisive (Top-Down)	All observations start in one cluster. The cluster is recursively divided into smaller clusters.

This experiment uses **agglomerative hierarchical clustering**, which is generally easier to implement and is the approach shown in the supplied college experiment.

4. Agglomerative Hierarchical Clustering

In agglomerative clustering, every data point initially forms a separate cluster. The algorithm computes distances between clusters, selects the closest pair, merges them, updates the cluster distances, and repeats the process. The procedure stops when the required number of clusters is reached or when the complete hierarchy has been constructed.

5. Distance / Proximity

A distance measure determines how similar or dissimilar two observations are. For numerical data, **Euclidean distance** is one of the most common choices.

For points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$:

$$d(P, Q) = \sqrt{[(x_1 - x_2)^2 + (y_1 - y_2)^2]}$$

A smaller distance means that two points are more similar in the selected feature space. The quality of clustering depends strongly on the choice and scaling of the features and on the distance measure.

6. Linkage Methods

Once individual points become clusters, the algorithm needs a rule for measuring the distance between two clusters. This rule is called **linkage**.

Linkage	Cluster distance
Single linkage	Minimum distance between any point in cluster A and any point in cluster B.
Complete linkage	Maximum distance between any point in cluster A and any point in cluster B.
Average linkage	Average of pairwise distances between points in the two clusters.
Ward's method	Merges clusters based on the increase in within-cluster variance; commonly used with numerical data.

The supplied experiment uses **single linkage**. Therefore, when comparing two clusters, only the closest pair of points across the two clusters determines their cluster distance.

HIERARCHICAL CLUSTERING — THEORY & ALGORITHM

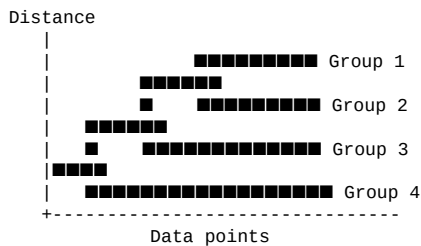
7. Single-Linkage Distance

$$d(A,B) = \min \{ d(p,q) : p \in A \text{ and } q \in B \}$$

For example, if cluster A contains A1 and A2 and cluster B contains B1 and B2, calculate all four cross-cluster distances. The smallest of the four values is the single-linkage distance between A and B.

8. Dendrogram

A **dendrogram** is a tree diagram that represents the sequence of merges in agglomerative clustering. Leaves represent individual observations. The height at which two branches join represents the distance at which their clusters were merged.



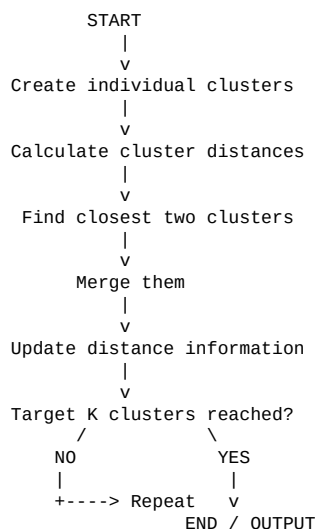
Cutting the dendrogram horizontally at a selected distance produces a particular number of clusters. This gives a visual way to study different possible clusterings.

9. Algorithm — Agglomerative Hierarchical Clustering

- Step 1: Read the dataset.
- Step 2: Treat every data point as one individual cluster.
- Step 3: Calculate the distance between every pair of clusters.
- Step 4: Find the two closest clusters.
- Step 5: Merge the closest clusters.
- Step 6: Update the cluster-distance information.
- Step 7: Repeat Steps 4-6 until the target number of clusters is reached.
- Step 8: Display the final clusters.

For single linkage, the updated distance between two clusters is the minimum Euclidean distance between any point in the first cluster and any point in the second cluster.

10. Algorithm Flow



11. Advantages

- Produces a complete hierarchy rather than only one clustering.
- A dendrogram makes the merging structure easy to inspect.
- The number of clusters does not have to be fixed before the hierarchy is constructed.

- Can be used with different distance and linkage measures.
- Useful for exploratory analysis when the underlying number of groups is unknown.

12. Limitations

- Naive implementations can be computationally expensive for large datasets.
- Once a merge is made, standard agglomerative clustering does not undo it.
- Results can change significantly with the choice of distance metric and linkage method.
- Different scales among features can distort distance calculations.
- Outliers can affect the resulting hierarchy.

13. Applications

- Customer segmentation based on age and spending behavior.
- Grouping documents or news articles.
- Image and pattern analysis.
- Biological and gene-expression analysis.
- Grouping products or users based on similarity.
- Exploratory analysis of high-dimensional datasets.

PART B — SAMPLE EXAMPLE

Consider the same type of dataset used in the college experiment. Each record contains **Age** and **Spending Score**. The objective is to group customers into **2 clusters** using agglomerative hierarchical clustering with Euclidean distance and single linkage.

Point	Age	Spending Score	Expected group
P1	18	85	Young high-spenders
P2	20	90	Young high-spenders
P3	22	80	Young high-spenders
P4	19	95	Young high-spenders
P5	55	20	Older low-spenders
P6	60	15	Older low-spenders
P7	50	25	Older low-spenders
P8	65	10	Older low-spenders

14. Step 1 — Initial Clusters

Initially, every point is treated as its own cluster:

```
C1 = {P1}
C2 = {P2}
C3 = {P3}
C4 = {P4}
C5 = {P5}
C6 = {P6}
C7 = {P7}
C8 = {P8}
```

15. Step 2 — Calculate Euclidean Distances

For example, the distance between P1 = (18,85) and P2 = (20,90) is:

```
d(P1,P2)
= √[(18-20)² + (85-90)²]
= √(4 + 25)
= √29
≈ 5.39
```

Some useful nearest-neighbor distances are:

Pair	Distance (approx.)
P1 – P2	5.39
P1 – P4	10.05
P2 – P4	5.10
P2 – P3	10.20
P3 – P4	15.13
P5 – P7	5.83
P5 – P6	7.07
P6 – P8	7.07
P5 – P8	14.14

16. Step 3 — First Merges

The closest pair among the points is P2 and P4, with distance approximately 5.10. Therefore, they are merged first.

```
C = {P2, P4}
```

The next closest relationships are then examined. Points P1 and P3 are close to this young-customer group, so the single-linkage process progressively combines the young points. Similarly, P5, P6, P7 and P8 form the older low-spender group.

17. Single-Linkage Example Between Clusters

Suppose one cluster contains P1 and P2 and another contains P3 and P4. Single linkage checks all cross-cluster distances:

```
d(P1,P3)
d(P1,P4)
d(P2,P3)
d(P2,P4)
```

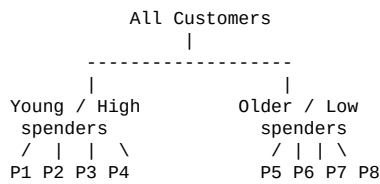
```
Single-linkage distance
= minimum of these four distances.
```

Thus, only the closest cross-cluster pair determines whether the two clusters should be merged next.

18. Final Two Clusters

Cluster	Members	Interpretation
Cluster 1	P1, P2, P3, P4	Young high-spenders
Cluster 2	P5, P6, P7, P8	Older low-spenders

19. Simplified Hierarchy



20. Result Interpretation

The hierarchical algorithm successfully separates the sample customers into two natural groups based on their Age and Spending Score. Customers P1–P4 have relatively low ages and high spending scores, while P5–P8 have higher ages and low spending scores.

The important point is that the algorithm did not receive the labels 'young high-spenders' and 'older low-spenders' during clustering. These descriptions are used only after examining the resulting groups. The algorithm itself works only from the feature values and their distances.

21. Result and Conclusion

Agglomerative hierarchical clustering was studied and implemented using Euclidean distance and single linkage. Each observation initially formed its own cluster, and the closest clusters were repeatedly merged until the required number of clusters was obtained.

The experiment demonstrates how hierarchical clustering can discover natural groups in unlabeled data and how a dendrogram can represent the complete sequence of cluster merges.